Knowledge that can change. Evidence that stays traceable.

Governed external-memory knowledge editing for connected factual questions.

Evidence phase	Independent lineages	Assigned generations	Execution
FINAL	32	1312	completed

The problem

An internal assistant can know yesterday's service owner and therefore name the wrong person to contact today. Updating one fact should change a connected answer while preserving other services, properties and historical releases.

The system we built

An operator approves versioned facts in a transactional memory store. Question-only retrieval supplies bounded current context to an attributed, upstream-trained ReasonKE model. The application records source facts and revisions, compares historical states, handles identity conflicts and supports withdrawal, restoration and portable text workspaces.

What is learned; what is deterministic

The pretrained neural reasoner generates answers and decides how supplied facts answer a question. Approval, canonical identity, version transitions and rollback are software controls. Publishing a fact does not modify weights. This project did not train the released ReasonKE or Qwen checkpoints.

This report is generated from saved model outputs, not published paper scores. Dataset review was automated, without independent human semantic review. The current study measures short factual answers in fictional handbooks, not arbitrary dialogue accuracy.

A controlled, reproducible question

Can an approved ownership change propagate to direct and two-hop answers without corrupting neighboring facts, and how do retrieval and the model checkpoint affect that behavior?

Ground truth and split

We independently authored 48 fictional service/team graphs with nine facts per world. An ownership edit changes service release 2; old and new teams retain their own contact facts. Each world has eight questions. The dataset contains 384 questions, split by lineage with seed 42 before wording augmentation: 16 development and 32 final worlds. Service, team and person names are disjoint across splits; simple values and templates can repeat.

This frozen final run executes all 32 held-out lineages. No final answers were used to choose prompts, retrieval, settings or aliases.

Question family	Assigned per main configuration
Direct changed fact	32
Direct paraphrase	32
Two-hop connected answer	32
Two-hop paraphrase	32
Same service, another property	32
Similarly named service	32
Older release	32
Unrelated fact	32

Paraphrases within a lineage are correlated. The independent unit is the edit lineage. The older-release category retains its machine key wrong_version_or_tier; this dataset actually tests release versions, not subscription tiers.

Five matched configurations

ID	Checkpoint and supplied information
R0	ReasonKE / initial facts
R-RAG	ReasonKE / updated RAG
R-ALL	ReasonKE / full memory
G-RAG	Qwen / updated RAG
G-ALL	Qwen / full memory

R0 receives initial facts and is scored against the new targets: it diagnoses stale knowledge. All updated conditions receive the same approved current information and supporting facts. Updated RAG replaces superseded values; it is not deliberately left stale. Both RAG generators receive identical selected fact IDs. Full-memory conditions expose all nine current facts.

Retrieval and generation

A two-stage lexical retriever first selects two question-relevant facts, expands the query with their answer text, and supplies up to four facts. Main retrieval never sees hidden target answers or gold support IDs. Oracle diagnostics use gold support only in their explicitly labelled smaller subset.

Both checkpoints have about 7.6 billion parameters and run sequentially on one Tesla T4 with NF4 double-quantized weights, FP16 compute, greedy decoding, a 4,096-token input cap and a 512-token output cap. ReasonKE uses its attributed updated-information prompt; Qwen uses its official chat template with short-answer instructions. Exact checkpoint ancestry is not verified, so this is not a causal ablation of upstream training.

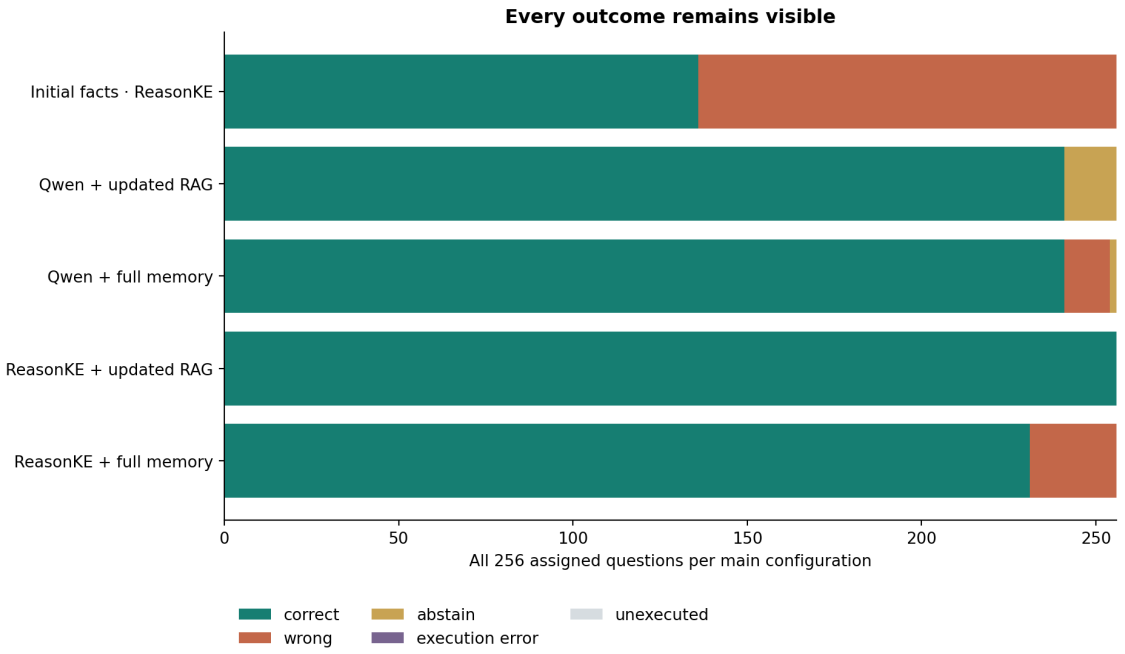
Finite execution and source control

Each request has a hard 240-second parent watchdog and loading has a 1,800-second watchdog. Receipts are saved atomically after every outcome. Resume runs only unexecuted records under identical source, configuration, data and protocol identities. Completed wrong/error answers are never retried for a better score.

The final protocol was frozen before inference. It assigns 784 ReasonKE and 528 Qwen requests: five 256-question main conditions plus two 16-question oracle-support diagnostics. The separate development study has 120 requests; profiles have 24 requests with different sampled mixes.

Measured answers, with all outcomes

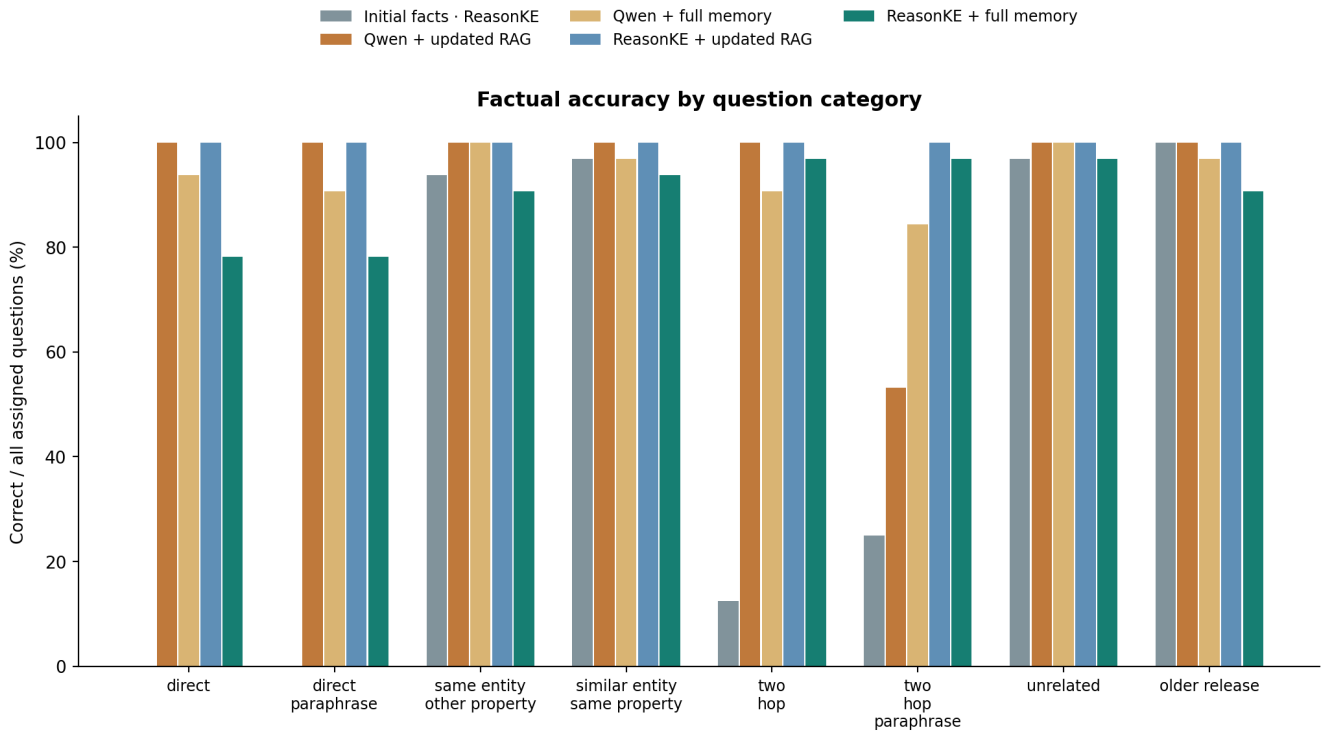
Condition	Correct	Wrong	Abstain	Error	Unexecuted	Coverage
R0	136/256	120	0	0	0	100.0%
R-RAG	256/256	0	0	0	0	100.0%
R-ALL	231/256	25	0	0	0	100.0%
G-RAG	241/256	0	15	0	0	94.1%
G-ALL	241/256	13	2	0	0	99.2%



Primary accuracy is correct divided by every assigned question. Abstentions, execution errors and unexecuted cases receive zero primary credit. Coverage is the fraction that gives a nonabstaining answer. Answered-only precision is separate; abstentions do not disappear from the success denominator.

Answers must match the whole endorsed final span against fixed aliases for the requested subject, relation and scope. Mentioning a target as a rejected alternative or inside an explanation is insufficient. The strict matcher can reject a semantically valid longer answer; the raw output remains available, without post-test alias changes.

Generalization and connected facts



32 assigned questions per category and configuration. Errors, abstentions and unexecuted cases receive zero primary credit.

Condition	Required facts retrieved	Complete two-hop support
R-RAG	320/320	64/64
G-RAG	320/320	64/64

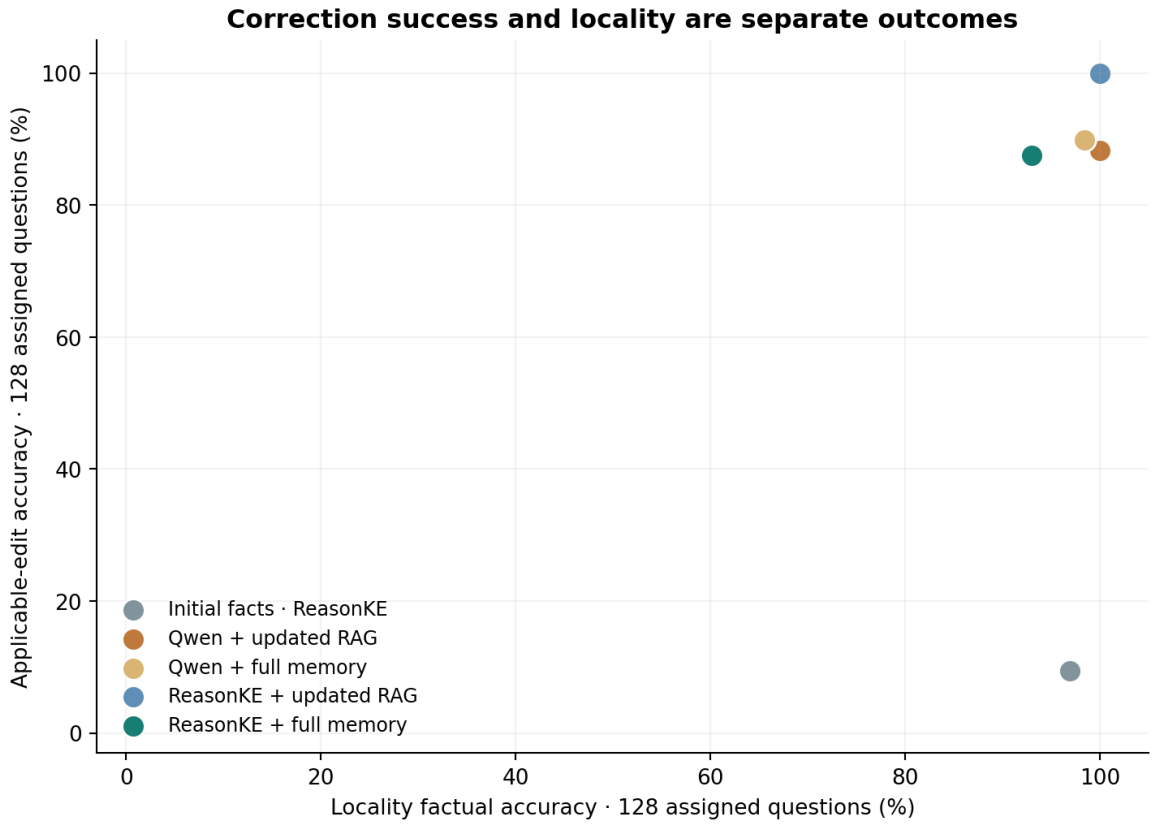
Retrieval success is not generation success: the model may receive both correct supporting facts and still connect the new owner to the old contact. The two models share the same retrieval procedure, so the two rows are not independent retrieval experiments.

R-ORACLE: 16/16 correct with oracle supporting facts on preselected two-hop questions.

G-ORACLE: 16/16 correct with oracle supporting facts on preselected two-hop questions.

Direct and two-hop paraphrases test unfamiliar wording within shared templates and unseen entities. They do not establish robustness to all natural-language formulations or long conversational chains.

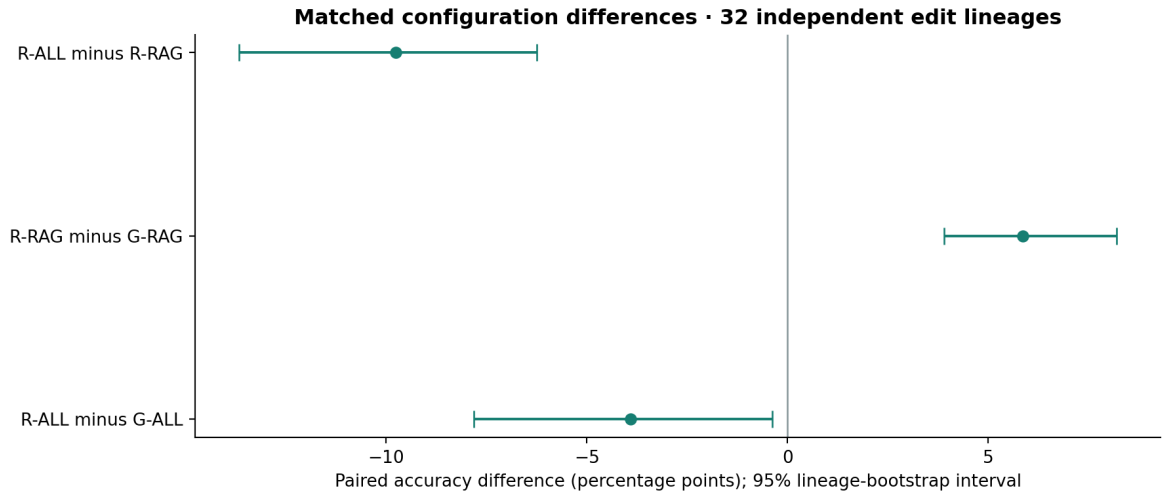
Preservation is not correctness



Condition	Locality correct	Retained R0-correct	False edited values
R-RAG	128/128	124/124	0/128
R-ALL	119/128	117/124	3/128

Locality accuracy checks truth on every out-of-scope question. Retention checks only those locality answers that R0 already answered correctly. Preserving an incorrect answer cannot count as factual success. False edited-value endorsements detect the corrected value being asserted for a question outside its scope; other wrong-answer types are reported separately.

Uncertainty and measured cost



Paired differences use the same questions and bootstrap whole lineages (10,000 resamples, seed 42). This study has 32 independent lineages; wording variants are not independent samples. A zero-width interval when two methods answer every sampled question correctly does not imply certainty outside this controlled sample.

Condition	Generated n	p50 seconds	p95 seconds
R0	256	15.01	18.17
R-RAG	256	15.06	16.71
R-ALL	256	15.35	18.72
G-RAG	256	0.92	1.20
G-ALL	256	1.44	1.66

Warm pipeline latency includes context selection, local worker communication and generation. It excludes cold loading and the website/browser round trip. The separate operational HTTP workload has a different latency definition.

reasonke: loading 79.63 s; peak reserved CUDA during loading 6.65 GiB. These are actual session measurements, not an unmeasured hardware guarantee.

reasonke: loading 237.81 s; peak reserved CUDA during loading 6.65 GiB. These are actual session measurements, not an unmeasured hardware guarantee.

instruction: loading 690.37 s; peak reserved CUDA during loading 6.65 GiB. These are actual session measurements, not an unmeasured hardware guarantee.

Failures, limits and concrete outcome

Case / condition	Expected	Actual / outcome
ops-005-two_hop_paraphrase / G-ALL	Nira Dane	Indigo unit 136 Nira Dane
ops-006-direct_paraphrase / G-ALL	Silver unit 77	Saffron unit 137
ops-016-direct_paraphrase / G-ALL	Saffron unit 87	Violet unit 147
ops-027-two_hop / G-ALL	Orin Lorne	Mara Arden
ops-027-two_hop_paraphrase / G-ALL	Orin Lorne	Mara Arden
ops-034-direct / G-ALL	Violet unit 105	Willow unit 165

Deterministic first-six failure subset in record-ID order. All failures remain in raw receipts. Absence of a failure in this subset is not evidence that it did not occur.

A correct final answer does not verify its generated explanation. Explanation support is audited separately; generated traces do not prove faithful internal reasoning.

The separate 32-trace final audit found 32/32 supported extracted claims with retrieval. Full memory had 26 supported, 3 contradicted and 3 unsupported extracted claims. Surrounding prose stays unassessed; no whole explanation is certified.

Concrete outcome and measured tradeoffs

What the product delivers

A working conversational editing interface with explicit fact approval, canonical scope identities, conflict resolution, version history, historical comparison, withdrawal/restoration, text-workspace export/restore, result exports and scientific graphs. Model output, deterministic state-machine checks and source context remain visibly distinct.

What the evidence supports

The selected ReasonKE retrieval configuration answered 256/256 held-out questions correctly: 128/128 applicable questions and 128/128 locality questions. Qwen with the same retrieved facts scored 241/256, with 15 abstentions and 0 wrong final spans. Every abstention remained in the denominator.

The paired retrieval accuracy difference was 5.86 percentage points (95% lineage-bootstrap interval 3.91 to 8.20). The cost was higher warm latency: ReasonKE retrieval p50 15.06 s versus Qwen 0.92 s. Under full memory, ReasonKE scored 231/256, versus Qwen 241/256; the specialized checkpoint was not best under every context policy.

The bounded objective was achieved by the selected configuration: changed facts propagated to direct and connected answers while locality answers stayed correct in this test. Context selection and the accuracy/coverage/latency tradeoff are the concrete experimental findings. The deployment policy was selected on development before these held-out answers. Shared templates, automated labels and strict short-answer matching limit the conclusion; the comparison does not isolate the causal effect of upstream training or establish arbitrary-language reliability.

Limits: shared wording templates, a small fictional population, automated labels, strict alias scoring, no calibrated neural confidence, identity-based rather than complete semantic conflict detection, nine-fact evaluated worlds, and no production multi-user authorization. Checkpoint training and research algorithms are upstream contributions.

Reproduction and evidence map

Artifact	Purpose
results.json	All aggregate denominators, retrieval, coverage, latency and paired uncertainty
Raw model receipts	Every assigned question, supplied facts, prompt hash, raw output, model revision and outcome
host-audit.json	Mechanical replay of identities, extraction and scoring; zero new model calls
derived/figure-manifest.json	Figure hashes, plotting source identity and input receipt hashes
configs/deployment-policy.json	Context policy selected from development before final evaluation
docs/OPERATIONS_DEMO.md	Website walkthrough, ownership change, two-hop behavior and rollback
docs/CHAT_API.md	HTTP schemas, scope rules and workspace restoration
notebooks/04 and 05	Pinned chat deployment and explicitly staged sequential-model evaluation

Source: github.com/ShivamRaghav7/verity. The inference source/data/configuration hashes are recorded in each receipt. Re-run scripts.audit_operations_evidence to recompute integrity and scoring; scripts.build_operations_report and scripts.build_operations_pdf regenerate exports without loading a model.

Attributed model checkpoints

YukinoKaren/ReasonKE_plus
revision ceaedc2ae7c8fed6257953587d991ce550db6106

YukinoKaren/ReasonKE_plus
revision ceaedc2ae7c8fed6257953587d991ce550db6106

Qwen/Qwen2.5-7B-Instruct
revision a09a35458c702b33eeacc393d103063234e8bc28

Reason-KE research: official code; Reason-KE++ paper. Published scores are not substituted for our measurements. ReasonKE checkpoint weight-license metadata was not established; weights are downloaded from the pinned upstream source, not redistributed in the website container.

Generated UTC: 2026-10-27T12:04:19.532396+00:00
Result SHA-256: 6bd7a1185644109982a5c7e636287616edb227ab4edac113d408d33a1c5effbe